

Humanoid Path Planner

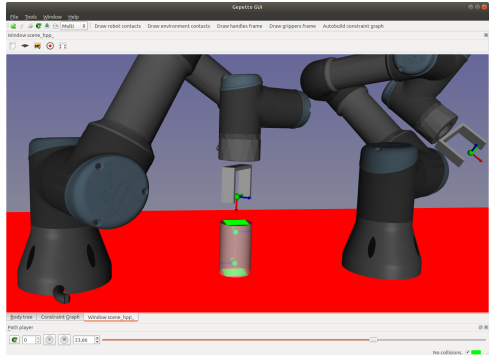
Florent Lamiraux

CNRS-LAAS, Toulouse, France

Definition of a manipulation planning problem

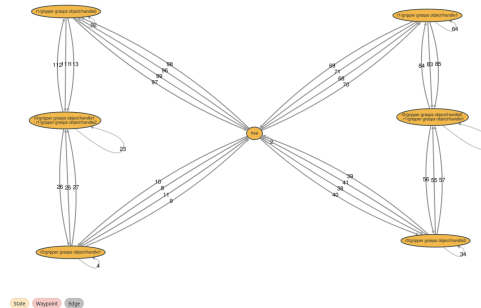
- ▶ grippers (frames attached to a joint),
- ▶ handles (frames attached to a joint),
- ▶ contact surfaces

define grasps and stable poses for objects.

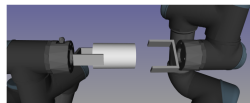
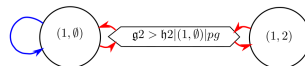


The factory

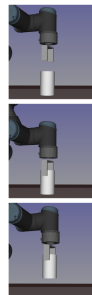
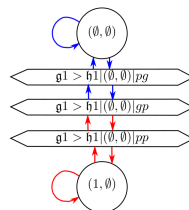
Given all the possible grasps (defined by grippers and handle), a factory builds the constraint graph with all possible states



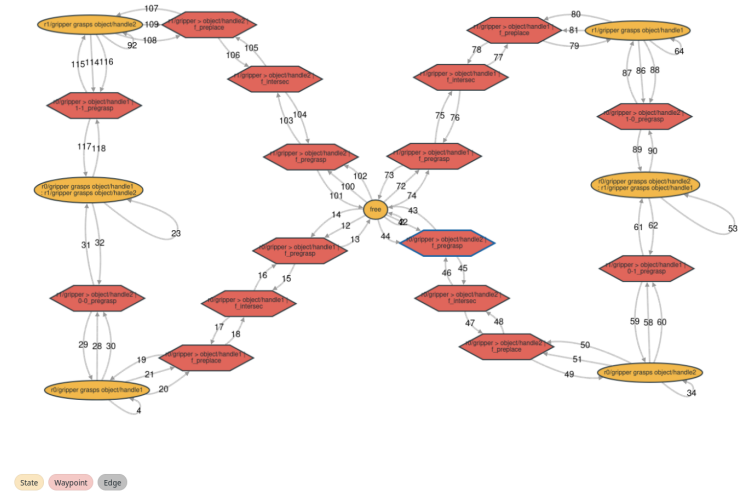
Waypoint transitions and states



To ease manipulation planning resolution, some intermediate states are inserted by the factory.



Waypoint transitions and states



Resources

- ▶ F. Lamiriaux and J. Mirabel, Prehensile Manipulation Planning: Modeling, Algorithms and Implementation, IEEE transactions on Robotics, 2022,
- ▶ Online documentation